



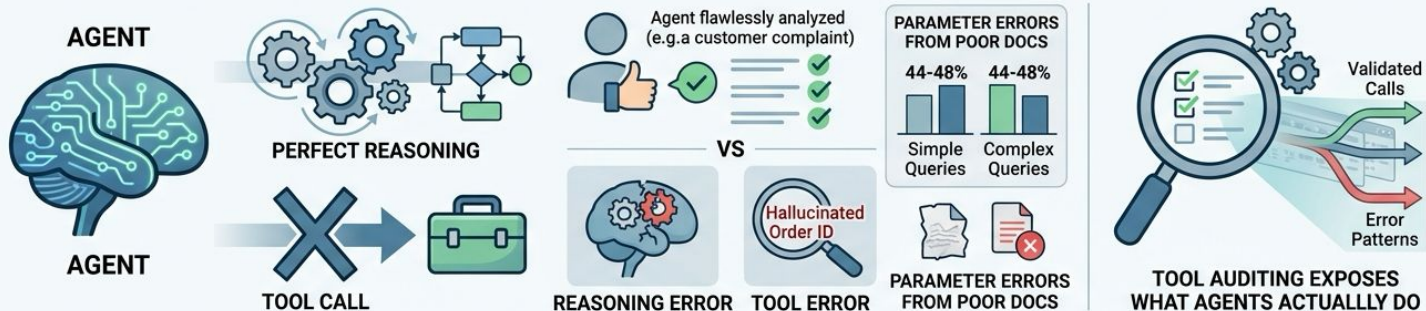
U.S. General Services
Administration

Chapter 3.7: Tool Usage Auditing

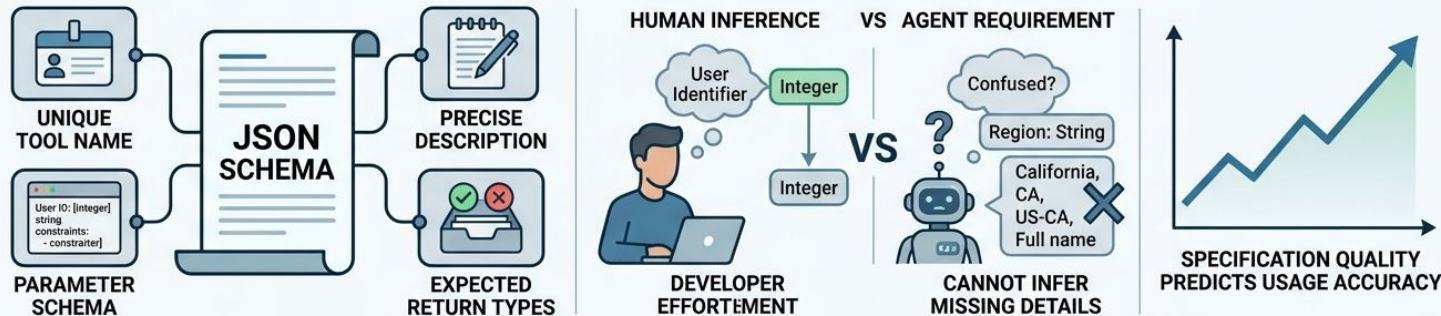
US General Service Administration
AI Community of Practice - March 2026

Tool Contracts Through Formal Specifications

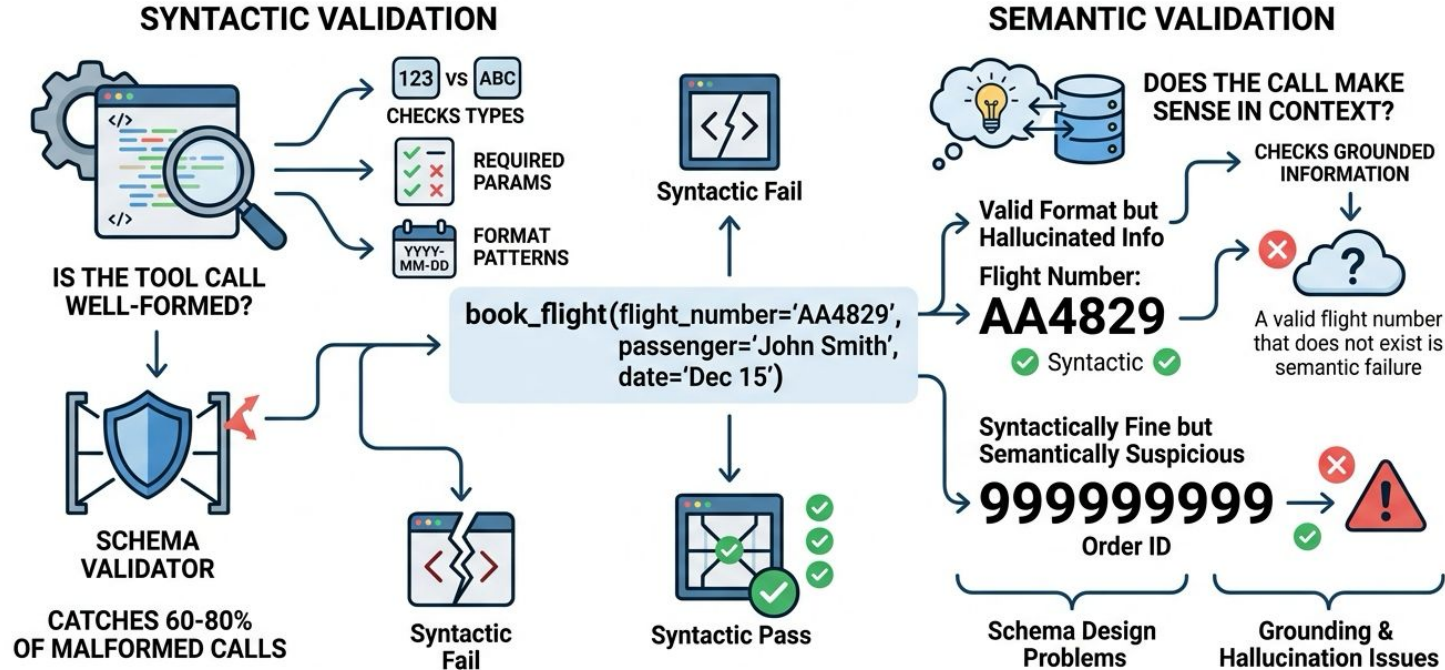
WHEN GOOD REASONING MEETS BAD TOOL CALLS



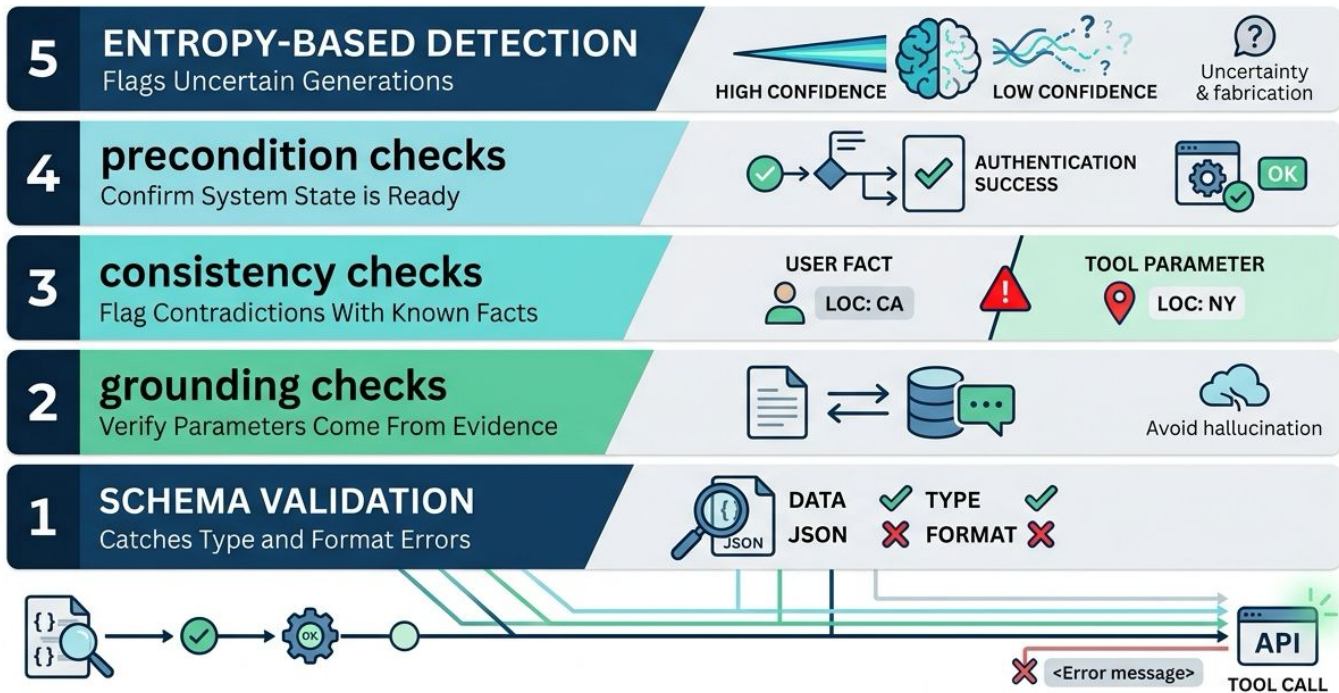
TOOL CONTRACTS THROUGH FORMAL SPECIFICATIONS



Syntactic vs. Semantic Validation

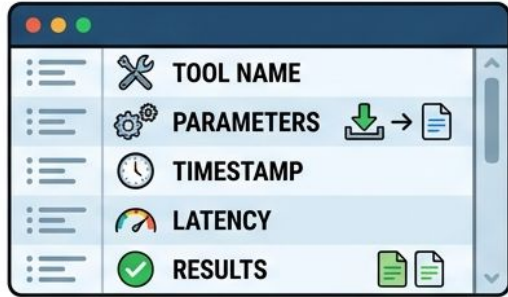


Pre-Execution Validation Layers



"PRE-EXECUTION AUDITING PREVENTS MOST TOOL ERRORS."

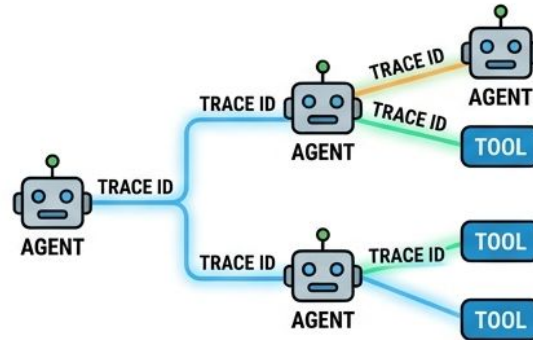
Post-Execution Monitoring



A screenshot of a web application interface for logging execution data. It features a sidebar with icons for each row: a wrench for 'TOOL NAME', a gear for 'PARAMETERS', a clock for 'TIMESTAMP', a rainbow for 'LATENCY', and a checkmark for 'RESULTS'. Each row has a corresponding icon and a download/upload icon.

TOOL NAME
PARAMETERS
TIMESTAMP
LATENCY
RESULTS

Comprehensive Execution Log



Distributed Tracing with OpenTelemetry

SCHEMA VALIDATION

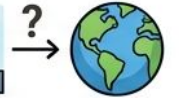


ENSURE THE
RESPONSE
STRUCTURE FITS
A TEMPLATE

SEMANTIC ANOMALY DETECTION

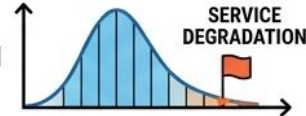


TOKYO-TOKYO



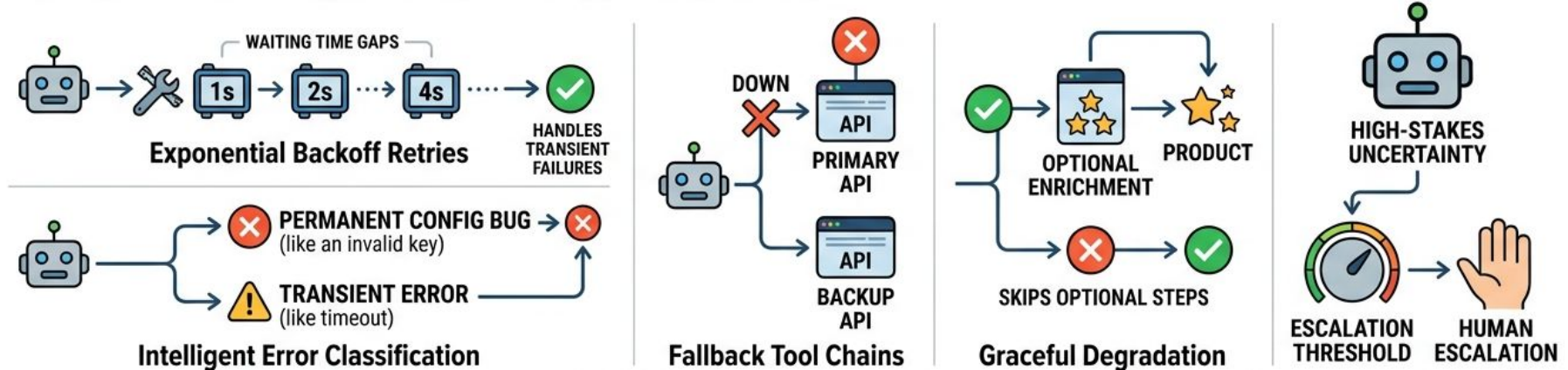
TOKYO-NEW YORK

LATENCY DISTRIBUTION TRACKING



- Log every invocation: tool, params, timestamp, latency, result
- Use OpenTelemetry for distributed tracing across agents
- Validate response schemas before downstream consumption
- Detect semantic anomalies in returned data
- Track latency distributions to spot service degradation

Recovery Mechanisms When Tools Fail



- Exponential backoff retries handle transient failures
- Distinguish transient errors from permanent configuration bugs
- Fallback tool chains provide alternative capabilities
- Graceful degradation skips optional enrichment steps
- Human escalation for high-stakes uncertainty

Tool Hallucinations

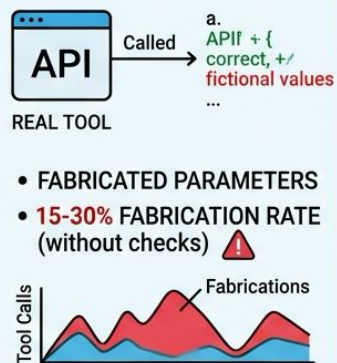
THE PROBLEM

PHANTOM TOOLS



⚠ AGENTS INVENT TOOLS

PARAMETER HALLUCINATION



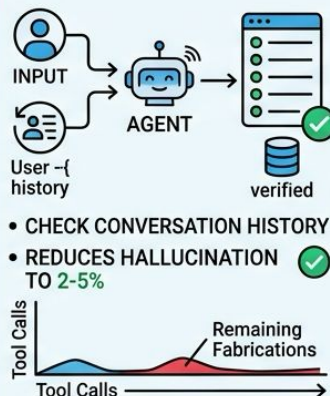
IMPROVE DOCUMENTATION



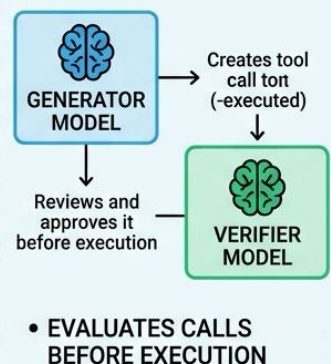
Eliminate your uncertainty to for your message

THE SOLUTIONS

GROUNDING VERIFICATION



TWO-MODEL ARCHITECTURE



ENTROPY ANALYSIS

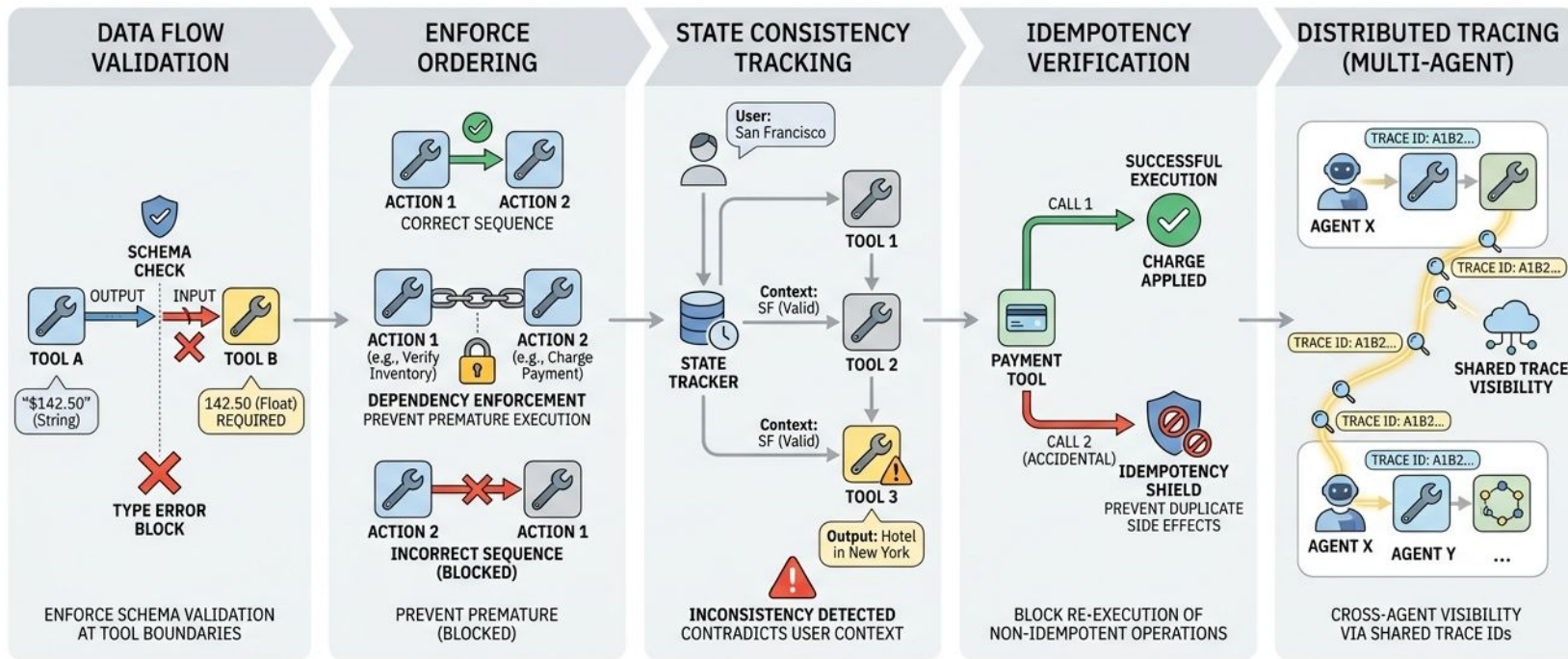


- FLAGS LOW-CONFIDENCE GENERATION

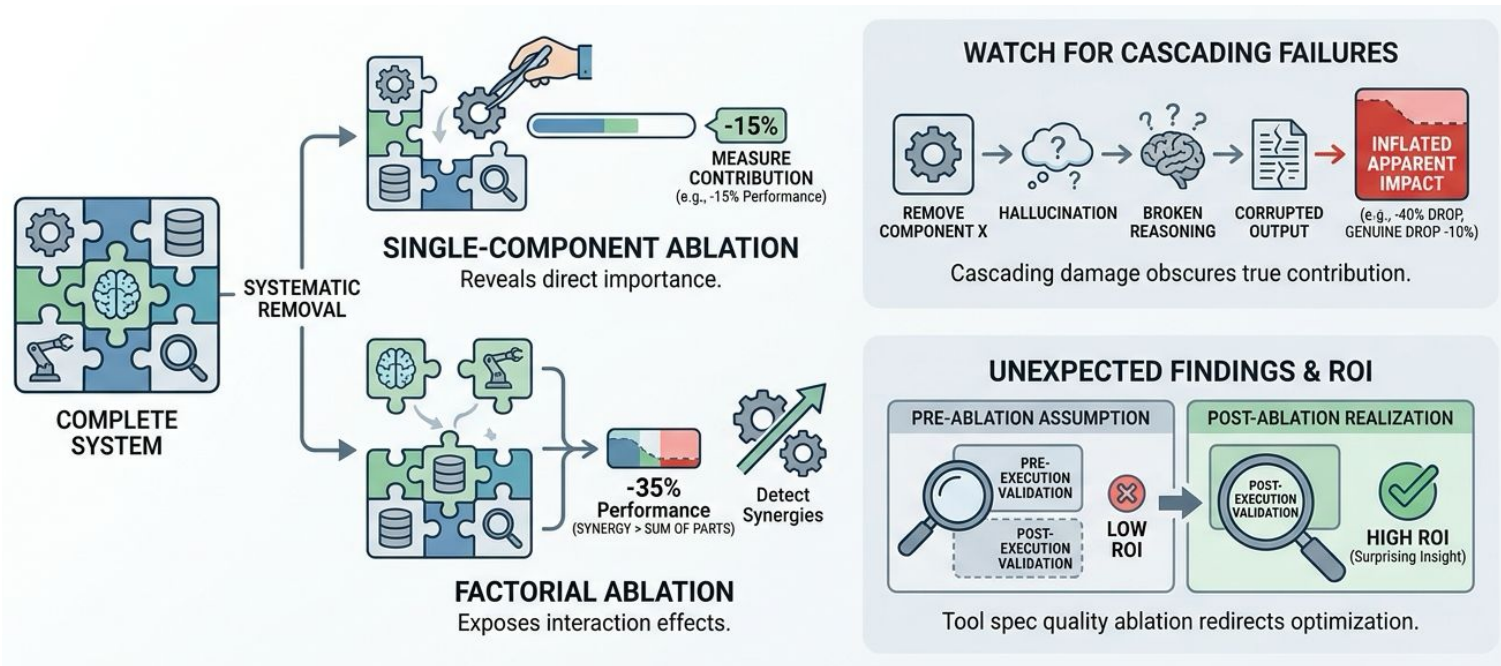
The Documentation-Accuracy Feedback Loop

- Vague tool descriptions cause tool selection errors
 - Missing parameter constraints cause format errors
 - Redundant tools without clear differentiation confuse agents
 - Standardize naming: `user_id`, not `uid`/`customer_id`/`userID`
 - Audit logs reveal which docs need improvement
- 

Auditing Multi-Step Tool Chains



Ablation Studies -- Finding What Actually Matters



Key Takeaways

- Tool auditing validates what agents do, not just what they think
 - Layer defenses: schema, grounding, consistency, preconditions
 - Documentation quality is the highest-leverage fix available
 - Audit logs drive continuous improvement cycles
 - Next: Chapter 3.8 measures action accuracy end to end
- 